Lab Experiment 2: Interactive Exploration of Tic-Tac-Toe using Reinforcement Learning

Task 1: Identify the Reinforcement Learning Components

RL Component	Student Observation
Agent	The student identified the AI player as the agent.
Environment	The student identified the Tic-Tac-Toe game board as the environment.
State	The student understood that the current arrangement of X's and O's represents the state.
Action	The student identified placing X or O in an empty cell as an action.
Reward	The student understood that the agent receives rewards based on winning, drawing, or losing.
Policy	The student understood that the policy determines which move the AI should choose.
Episode	The student understood that one complete Tic-Tac-Toe game represents an episode.]

Task 1 Analysis Questions

Q1. Who is the learning agent in this environment?

Answer: The learning agent is the Tic-Tac-Toe AI player. It learns from the game experience and selects the best possible move based on the values it has learned.

Q2. What is considered the environment?

Answer: The 3×3 Tic-Tac-Toe board and its game rules form the environment. It determines valid moves, updates the board, and identifies whether the game is won, lost, or drawn.

Q3. How is the game state represented?

Answer: The game state is represented by the current arrangement of X, O, and empty spaces on the 3×3 board. Each unique board arrangement represents a different state.

Q4. What are the possible actions available to the agent?

Answer: The agent can choose any empty cell on the board and place its assigned mark, either X or O.

Q5. When does the agent receive a positive reward?

Answer: The agent receives a positive reward (+1) when it wins the game by placing three of its marks in a row horizontally, vertically, or diagonally.

Q6. Which Reinforcement Learning approach is used for learning?

Answer: The system uses Temporal-Difference (TD) learning with self-play and an ε-greedy policy. The agent learns by trying different actions and updating its values based on the results.

Q7. Is the agent trained using labelled data? Justify your answer.

Answer: No. The agent does not use labelled training data or predefined correct moves. It learns through trial and error, playing games and improving its decisions based on the rewards it receives.

Q8. How does Reinforcement Learning differ from Supervised Learning in this example?

Answer: In Supervised Learning, the model learns from labelled examples containing the correct move. In Reinforcement Learning, the Tic-Tac-Toe agent learns through trial and error, chooses its own actions, receives rewards or penalties, and improves its strategy over time.

Task 2: Effect of Training Episodes on Agent Behaviour

Training Episodes	Win %	Loss %	Draw %	Behaviour Observed
100	Low	High	Very Low	The agent makes mostly random moves, misses simple winning chances, and does not respond effectively to the opponent's threats.
500	Moderate	Moderate	Low	The agent starts identifying some immediate winning opportunities and occasionally blocks threats, but its overall strategy is still inconsistent.
1,000	High	Low	Moderate	The agent shows noticeable improvement by blocking direct threats and selecting strategically useful center and corner positions.
5,000	High / Near-Optimal	Very Low	High	The agent demonstrates strong strategic behaviour, creates forks, avoids common traps, and frequently forces draws against strong opponents.

10,000	Optimal	~0%	Dominant	The agent demonstrates almost perfect play, effectively prevents losing situations, and consistently achieves draws when facing optimal play.

Task 2 – Analysis Questions

Q9. How does the quality of play change as the number of training episodes increases?
Answer: As training increases, the agent gradually develops better decision-making skills. Initially, its moves are mostly random, but with more experience it learns to block threats, occupy valuable positions, create winning opportunities, and finally approach optimal Tic-Tac-Toe play. Improvement becomes slower once the agent has learned most important strategies.

Q10. At approximately how many training episodes does the agent begin making intelligent decisions?
Answer: The agent begins showing clearly intelligent behaviour at around 1,000 episodes. At this stage, it can recognize immediate threats, block the opponent's winning moves, and make better choices involving the center and corner positions.

Q11. At what stage does the learning appear to converge?
Answer: The learning appears to converge between approximately 5,000 and 10,000 episodes. During this stage, the agent's behaviour becomes stable, and further training produces only small improvements in its decisions and performance.

Q12. Why does the win percentage improve with additional training?
Answer: More training episodes provide the agent with greater experience of different game situations. Through repeated interaction and value updates, the agent learns which actions lead to better outcomes. As the learned state values become more accurate, the agent makes increasingly effective moves.

Q13. Why do most games end in a draw after sufficient training?
Answer: Tic-Tac-Toe is a solved game in which perfect play from both players results in a draw. After sufficient training, the agents learn to avoid major mistakes and defend against winning opportunities. Therefore, when both agents play optimally, neither player can force a win, resulting in mostly draws.

Q14. What happens if the number of training episodes is too small?
Answer: With insufficient training, the agent does not gain enough experience to learn the important patterns of the game. Its state-value estimates remain inaccurate, causing it to make poor moves, miss winning opportunities, and fail to block the opponent effectively. As a result, the agent experiences more losses and unpredictable behaviour.

Analysis

The learning rate controls **how strongly new experiences change the agent's existing knowledge**. A very low learning rate results in slow learning, while a very high learning rate can make learning unstable. In this experiment, **α = 0.5** provides a suitable balance between learning speed and stability. Therefore, it was selected as the default learning rate for further training.

Analysis

The exploration rate determines **how often the agent chooses random actions instead of the action with the highest learned value**. A low ε allows the agent to make better use of its learned knowledge, while a high ε encourages excessive random exploration. In this experiment, **ε = 0.05** produced the most consistent final behaviour when combined with **α = 0.5**.

Gameplay Analysis

After training the agents for **10,000 episodes** using **α = 0.5** and **ε = 0.05**, the trained agent was evaluated through direct gameplay against a human player.

Game 1: Human Win Scenario

Game 1 – Human player successfully completes a winning line.

Game 1 Analysis

In this game, the human player successfully created a three-in-a-row winning combination. Although the agent had undergone extensive training, some less frequently encountered game states may not have been explored sufficiently during self-play. The value-function display showed relatively uncertain values for some states, such as **50% and 25%**, suggesting that these particular branches had limited experience during training. This demonstrates that even a well-trained agent can have uncertainty in rarely visited states.

Game 2: Draw Scenario

Game 2 – The game ends in a draw with the board completely filled.

Game 2 Analysis

In this game, neither player was able to create a three-in-a-row combination, resulting in a draw. The trained agent successfully identified and blocked the human player's immediate winning opportunities while selecting safe moves throughout the game. This behaviour demonstrates the improvement achieved through repeated self-play training. The draw is also consistent with the theoretical outcome of Tic-Tac-Toe when both players avoid tactical mistakes and follow optimal defensive strategies.

Overall Observation

The hyperparameter experiments show that both **learning rate (α)** and **exploration rate (ε)** have a significant effect on the agent's learning behaviour. Very small values result in slow learning, whereas very large values can lead to unstable or excessively random behaviour. Among the tested combinations, **α = 0.5 and ε = 0.05** provided a good balance between exploration, exploitation, learning speed, and stability. After **10,000 training episodes**, the agent demonstrated strong gameplay, although occasional uncertainty may remain in rarely visited states.